TRIASHA SARKAR

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Machine learning engineer with about two years of industry experience and graduate research experience in aerospace. Builds models and the systems around them, from data preparation and evaluation through APIs and deployment checks.

PROFESSIONAL EXPERIENCE

Graduate Research Assistant under Prof. Dimitri Mavris | Georgia Tech ASDL | May 2025 – Aug 2026

- Built and tested Python models for projects in simulation, airline safety, and sustainable aviation; shared reproducible analyses with faculty and sponsors.
- Reviewed assumptions and limits with aerospace experts and documented each workflow so other researchers could rerun it.

Machine Learning Engineer | Rolls-Royce | Jul 2023 – Apr 2025

- Developed Python pipelines to flag unusual engine behavior, support maintenance planning, and combine sensor streams sampled at different rates.
- Mapped certification requirements to data and model checks, then reviewed results with lifecycle engineers before use.

Data Science Intern | Rolls-Royce DataLabs | May 2021 – Jul 2021

- Cleaned aircraft-engine sensor data, built features, and compared prediction and anomaly models; summarized recurring failure patterns for engineers.

TECHNICAL PROJECTS

EdgeGenBench | Real-flight ML and on-device inference | Aug 2026 – Present

- Trained a flight-anomaly model on NASA DASHlink recordings and tested it on 17,780 approaches from held-out aircraft. Its 0.738 macro F1 missed the release target.
- Exported the model to ONNX and tested corrupted sensor inputs; prediction agreement stayed above 99.55% across the tested cases.

IntegrityBench | Moderation evaluation and release controls | Aug 2026 – Present

- Built controlled evaluation and release checks for changing moderation policies, keeping separate candidates and frozen benchmark results distinct from deployment claims.

AeroSynth-Eval | Synthetic inspection-image evaluation | Aug 2026 – Present

- Evaluated synthetic aircraft-inspection image data with reproducible splits; reported where results were strong, limited, or still pending validation.

NewsLens | Recommendation and real-time search | Jul 2026 – Present

- Used chronological splits to keep future clicks out of training. The recommender scored 0.366 NDCG@10, but the uncertainty interval still included no improvement.
- Built a Go, Kafka, PostgreSQL, and FastAPI ingestion path for new articles. A 60-minute local soak accepted 100,000 events with 0% publish failures, 224 ms freshness p95, and 3.1 s outage recovery; multi-host and online product impact remain unmeasured.

Equity Backtest | Walk-forward signal evaluation | Jul 2026 – Sep 2026

- Built a cross-sectional equity-signal backtest with trading costs and safeguards against common look-ahead and selection biases.

Atlanta Mobility Resilience Digital Twin | Independent project | 2026 – Present

- Built an OSMnx and MARTA GTFS graph-analysis workflow covering 50 Census-tract origins, 101 essential-service destinations, and 216,659 estimated residents; the planned accessibility validation cycle is still pending.

Silent Failure Detection in Rocket-Motor Simulations | AE8900 Special Problems Coursework | May 2026 – Jul 2026

- Traced silent failures in a 15,120-case rocket-motor study to an uncapped convergence loop; a leakage-safe classifier reached 96.8% accuracy on unseen geometries.

Surrogate Model Learning | Reliability under design shift | Feb 2026 – Oct 2026

- Evaluated Gaussian-process and conventional surrogates on public airfoil, energy, and concrete data using grouped splits, multi-seed analysis, split-conformal intervals, and distance-to-training-domain guards.
- An airfoil Gaussian process reached R^2 0.8145 on a physically grouped split; a 10-seed mean of 0.8662 ± 0.0680 exposed split sensitivity. Nominal 90% intervals did not retain 90% coverage under design shift.

AIRFAANS | Extension of AE6394 Coursework Project | Jan 2026 – Present

- Compared an MLP, MeshGraphNet-style GNN, and point neural operator across three matched seeds and 200 AirfRANS interpolation meshes per model; MeshGraphNet had the lowest mean field and drag error.

- Completed six Reynolds-OOD treatments and a scarce-data replication; AoA-OOD, uncertainty calibration, active learning, and operational readiness remain pending.

AeroRAG-X | Independent extension of Project HERO | Retrieval, evaluation, and ML systems | Oct 2025 – Present

- Extended Project HERO independently into a technical-knowledge system over 3,233 NASA report sections using hybrid retrieval, reranking, evidence checks, controlled citations, and containerized FastAPI services; retained negative experimental findings rather than claiming every change helped.

Project EAGLE | RIC Enterprise-sponsored Vehicle Grand Challenge | Georgia Tech ASDL | Sep 2025 – May 2026

- Developed sizing and synthesis tooling, passenger layout, AutoCAD cabin geometry, and requirements for the Eco-friendly Airborne Global Luxury Ekranoplan.

Project GREEN TEA | U.S. Endowment-sponsored Systems of Systems Grand Challenge | Georgia Tech ASDL | Sep 2025 – Present

- Developed a techno-economic framework for forest-residue sustainable aviation fuel and bioenergy, including jobs, CO2 impacts, a GREET surrogate, and forest-health modeling.

Project HERO | Delta Air Lines-sponsored source-aware retrieval for airline safety | Georgia Tech ASDL | May 2025 – Aug 2026

- Developed source-aware retrieval and evaluation methods for airline-safety analysis.

Supersonic Business Jets and Noise Mitigation | Sponsored by Rolls-Royce, Seneca, and DLR | Cranfield University | Sep 2022 – Feb 2023

Studied geometry, noise mitigation, high-speed aerodynamics, and design trade-offs.

Thermal System Design Engineer Group Project | Sponsored by GKN Aerospace | Cranfield University | Sep 2022 – Feb 2023

Worked in a multidisciplinary team on thermal-system architecture, requirements, sizing, and design trade-offs for an aerospace application.

Bird-Strike Impact Analysis | SRM Institute of Science and Technology | Jun 2020 – Jun 2021

Used ANSYS to compare aircraft nose-cone material choices for bird-strike impact.

Ornithopter Aerodynamics | SRM Institute of Science and Technology | Sep 2019 – May 2020

Used CFD to examine ornithopter wing area and load factor.

TECHNICAL SKILLS

ML and scientific ML: PyTorch, PyTorch Geometric, scikit-learn, graph neural networks, neural operators, surrogate modeling, uncertainty quantification, Transformers, RAG, BM25, dense retrieval, reranking

Systems: Python, Go, SQL, Kafka, FastAPI, Docker, Kubernetes, GitHub Actions, PostgreSQL/pgvector, Prometheus, OpenTelemetry, ONNX, Core ML, Qualcomm QNN

Scientific computing: NumPy, SciPy, pandas, CFD meshes, NetworkX, OSMnx, statistical inference, design of experiments

LEADERSHIP AND HACKATHONS

- Graduate Chair | Women of Aeronautics and Astronautics, Georgia Tech | Aug 2025 – Aug 2026
- Organising Team Member | Aarush Hackathon Event, SRM University | Jun 2019 – Jul 2019
- Student Volunteer | Aarush, SRM University | Aug 2018 – Jan 2019
- Team Participant | 36-Hour SRM Hackathon, SRM University | Sep 2019

  Worked in a five-member team on a sustainable urban-transport concept for densely populated Indian cities, comparing travel time, fuel use, cost, and CO2 emissions.

EDUCATION

M.S., Aerospace Engineering | Georgia Institute of Technology, Atlanta, GA | Aug 2026

GPA: 3.7/4.0

M.Sc., Aerospace Vehicle Design | Cranfield University, United Kingdom | Feb 2023

GPA: 3.6/4.0 | First Class Honours | Thesis: geometry optimization under FAA constraints

B.Tech., Aerospace Engineering | SRM Institute of Science and Technology, India | Jun 2021

GPA: 3.8/4.0 | First Class Honours